

How is this possible?

AI can now create breathtaking visuals from simple text descriptions. This capability represents a fundamental shift in creative expression and machine intelligence.

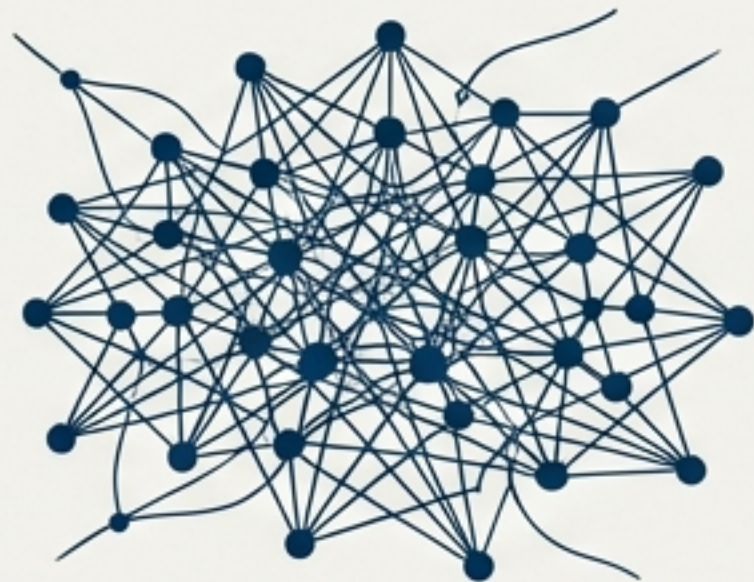
But this isn't magic. It's the result of two key breakthroughs working in concert.

This presentation explains how.



The Old Way: Computer Vision's Three Major Hurdles

For all its success, traditional deep learning for computer vision faced persistent challenges that limited its scope and reliability.



Costly & Laborious Datasets

Vision models have traditionally been trained on manually labeled datasets that are expensive to construct.

SUPPORTING FACT

The ImageNet dataset required over **25,000 workers** to annotate 14 million images for 22,000 object categories.

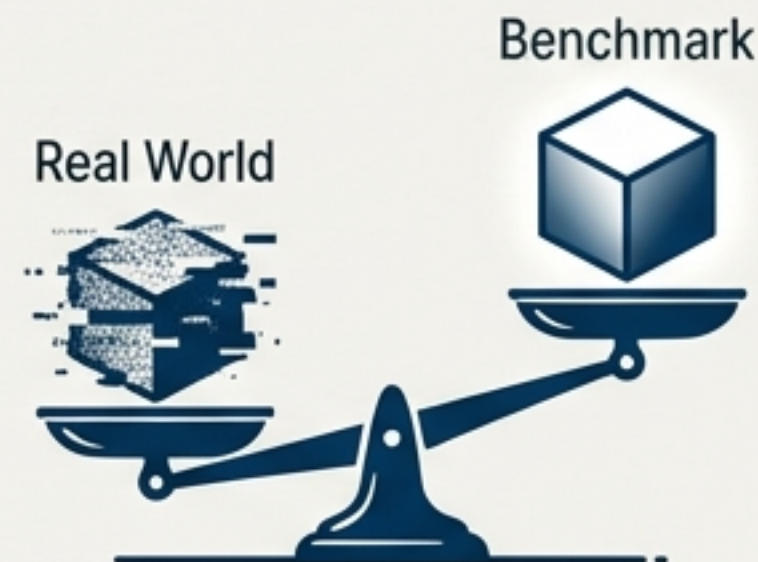


Narrow & Inflexible Models

A standard model is good at one task and one task only. Adapting it requires a new dataset, a new output head, and fine-tuning.

EXAMPLE

An ImageNet model can predict 1000 categories but **can't perform any other task "out of the box"**.



Brittle Real-World Performance

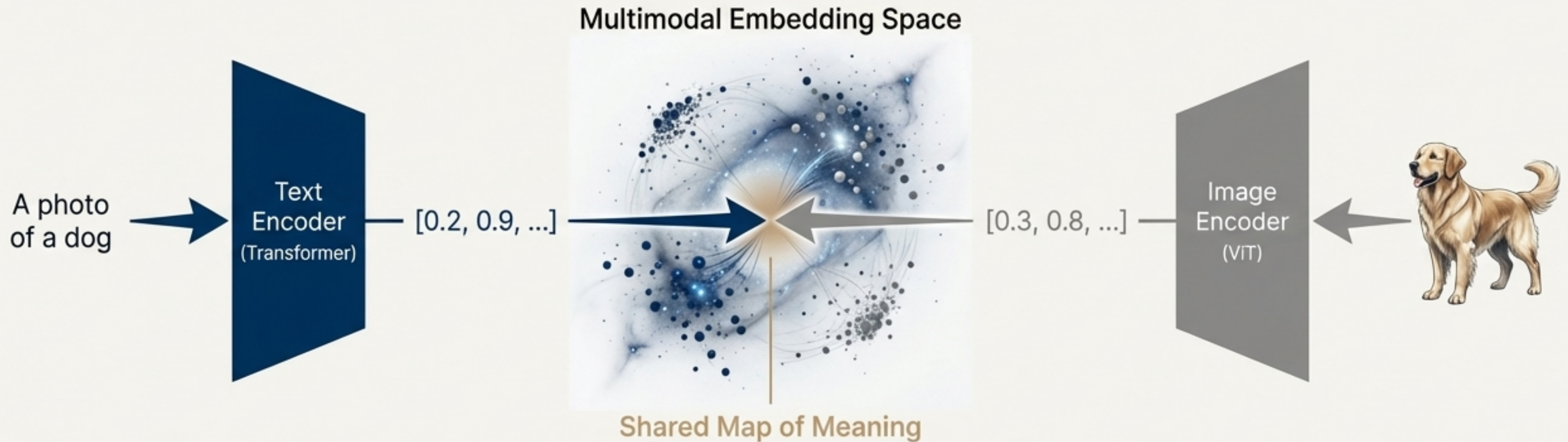
Models that excel on benchmarks often have disappointingly poor performance on stress tests, creating a gap between "benchmark performance" and "real performance".

CONCEPT

This happens because models **"cheat"** by optimizing for **the benchmark**, not for **general understanding**.

The Breakthrough: Learning from Language Itself

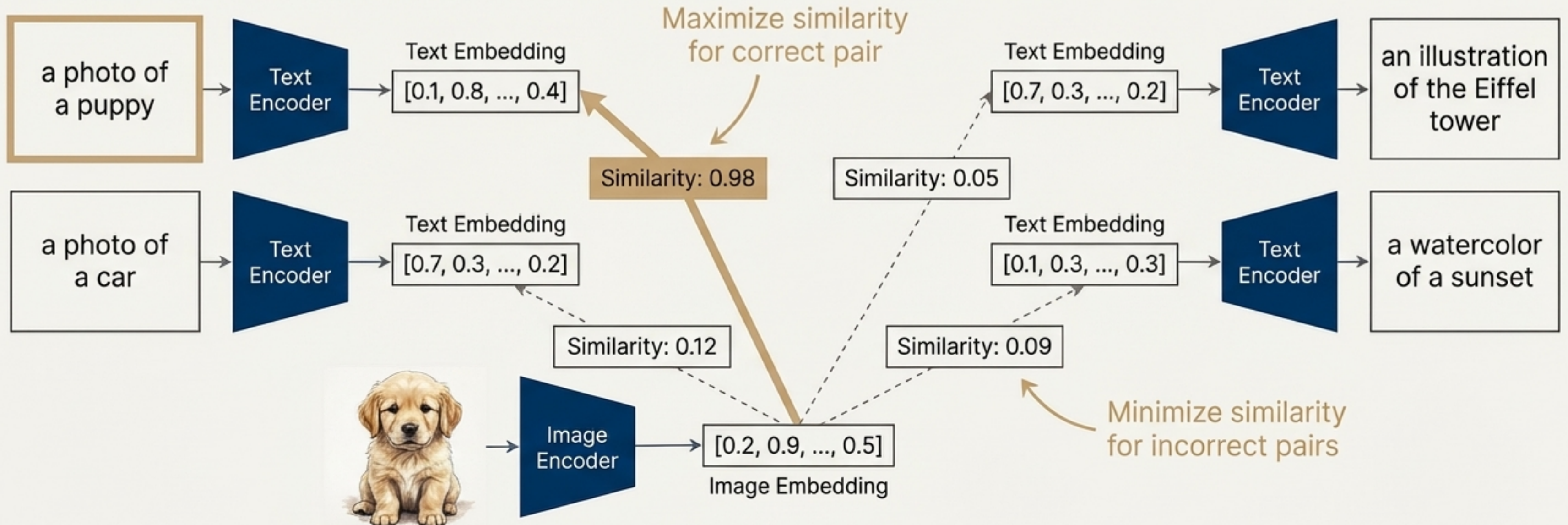
OpenAI introduced a neural network called CLIP (*Contrastive Language–Image Pre-training*) to address these problems. The core idea is to learn visual concepts not from manually curated labels, but from the vast amount of natural language supervision available on the internet.



Key Idea: CLIP learns to build a shared map of meaning where similar concepts, whether expressed in words or pixels, are located close together.

How CLIP Learns: A Game of Matching

CLIP is not trained to predict a specific label. Instead, its proxy task is: "Given an image, predict which out of a set of 32,768 randomly sampled text snippets was actually paired with it."



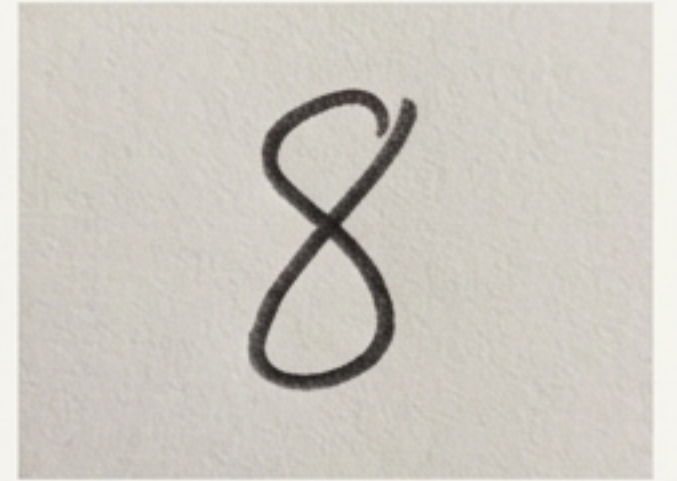
To succeed at this task, the model must learn to recognize a wide variety of visual concepts and associate them with their names. It learns to connect pixels to meaning.

The Superpower: True “Zero-Shot” Generalization

Because CLIP learns concepts from language, it can be applied to almost any visual classification task without any additional training.

How it Works: To perform a new task, you simply provide the text-encoder with the names of the visual concepts. For a “dog vs. cat” classifier, you ask CLIP whether the text “a photo of a dog” or “a photo of a cat” is more similar to the image’s embedding.

Without using any of the 1.28M labeled examples, CLIP matches the performance of the original, fully supervised ResNet-50 on ImageNet.



OCR: “8”



Geo-localization:
“Louvre Museum”



Action Recognition:
“swimming”

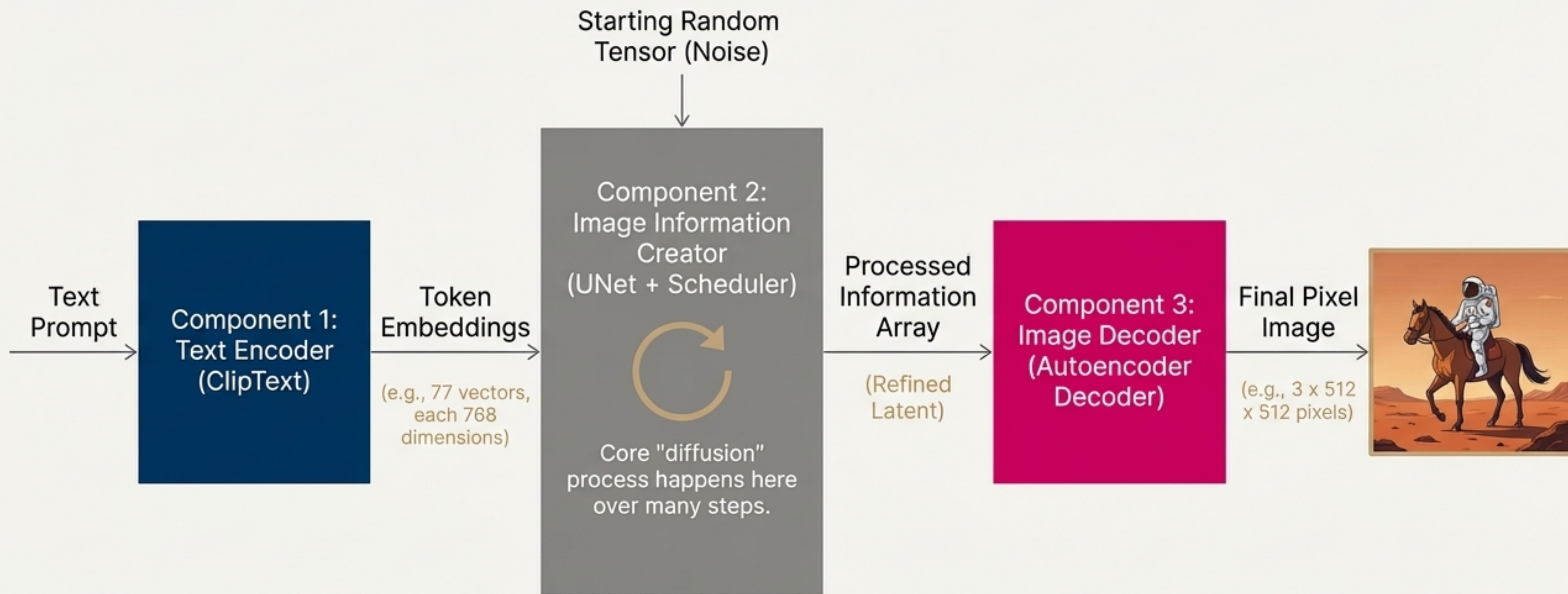
From Understanding to Creating

We've built a bridge between words and images. Now, can we use that bridge to build new worlds from words alone?

This is the role of **Stable Diffusion**. It is not one single model, but a system of components that leverages a CLIP-like text encoder to guide a powerful image generation process.



The Architecture of Stable Diffusion



The Core Idea: Generating by Denoising

Forward Process (Training): We start with a clean image and systematically add noise. The model's task is to learn how to predict the noise that was added at each step.

Training: Learning to predict added noise



Inter



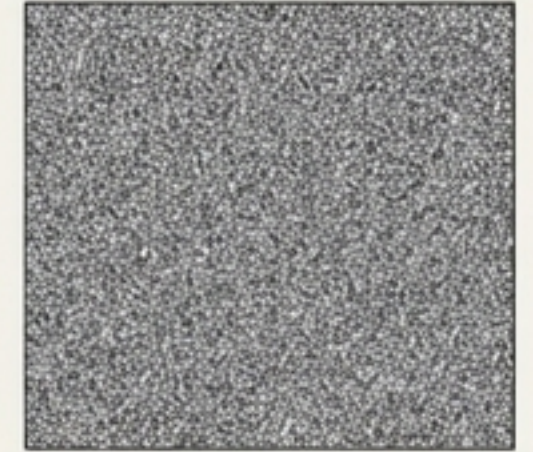
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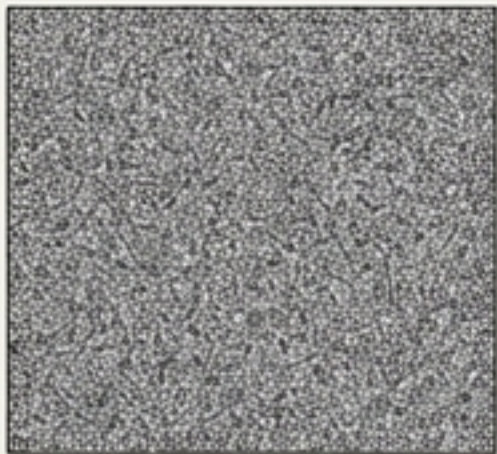


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Reverse Process (Inference): To generate an image, we start with pure random noise. We then use the trained model to iteratively subtract its noise prediction, step-by-step, revealing a coherent image.



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Inference: Iteratively subtracting predicted noise

The Sculptor's Process: An Image Emerges from Noise

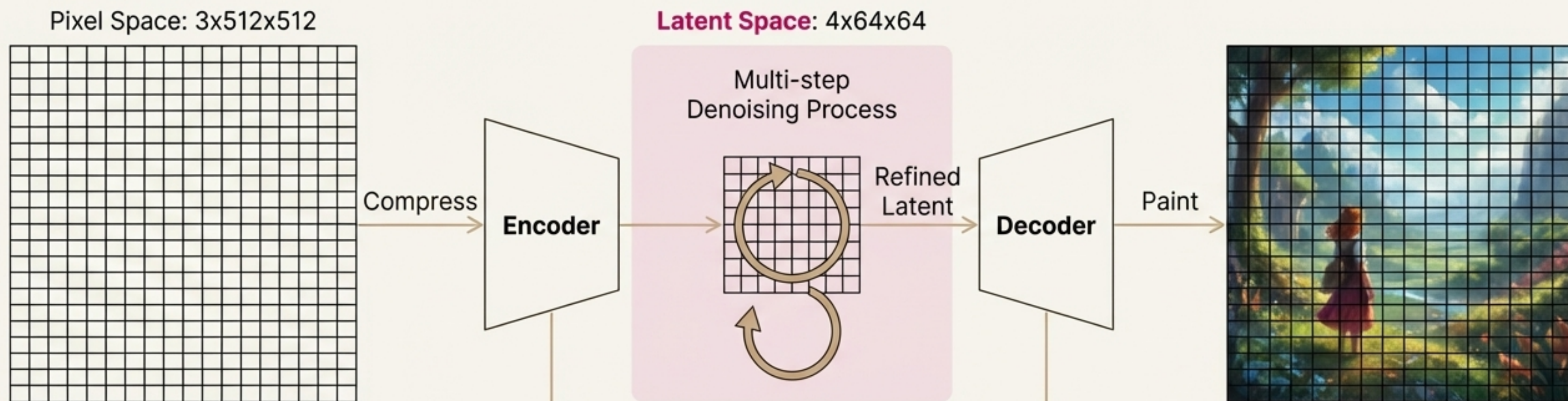
The “noise predictor” isn’t just removing random noise. It is **sculpting the noise** into a coherent image that resembles the *distribution* of images it was trained on—images where skies are blue, people have two eyes, and cats look unimpressed.

Something especially fascinating happens here.
It's as if the outline emerges from the noise.



The Speed Boost: Diffusion in a Compressed “Latent Space”

Running the diffusion process on full-size pixel images is computationally massive. Stable Diffusion’s key innovation is to first compress the image into a much smaller “latent space” using an **autoencoder**. The entire denoising process happens on this small, compressed representation.



Autoencoder's Role

Encoder:

Compresses a full-size image into a smaller latent representation. This happens during training.

Decoder:

Takes the final, denoised latent representation and **“paints”** it back into a full-size pixel image. This step runs only once at the end of generation.

Connecting Text to Creation: Guiding the Denoising

The Mechanism

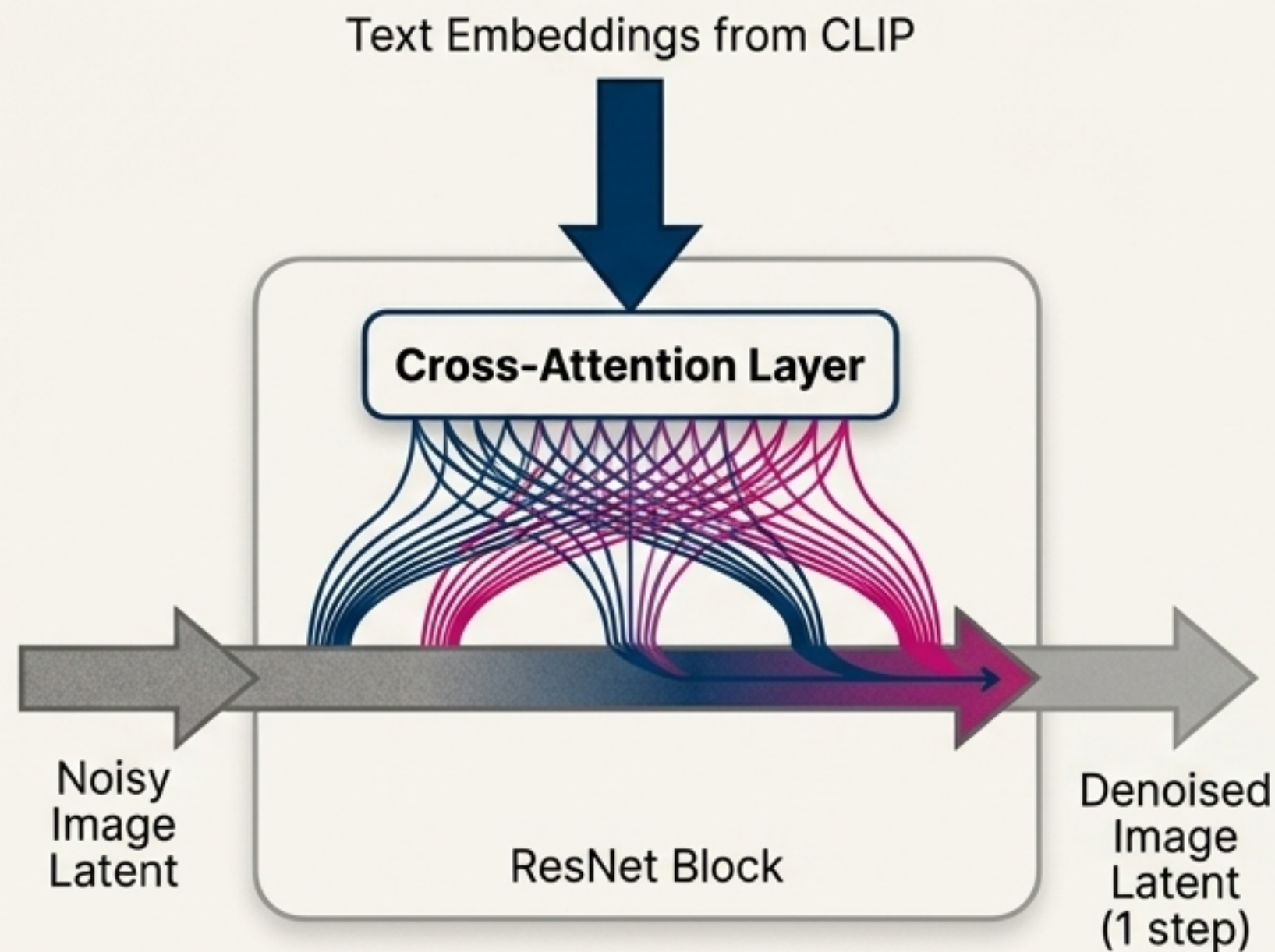
The noise predictor (UNet) doesn't just look at the noisy latent image. At each denoising step, it also looks at the text embeddings provided by the CLIP text encoder.

How it works inside the UNet

The UNet architecture is interspersed with **cross-attention layers**. These layers allow the model to merge the information from the text embeddings into its visual processing.

Analogy

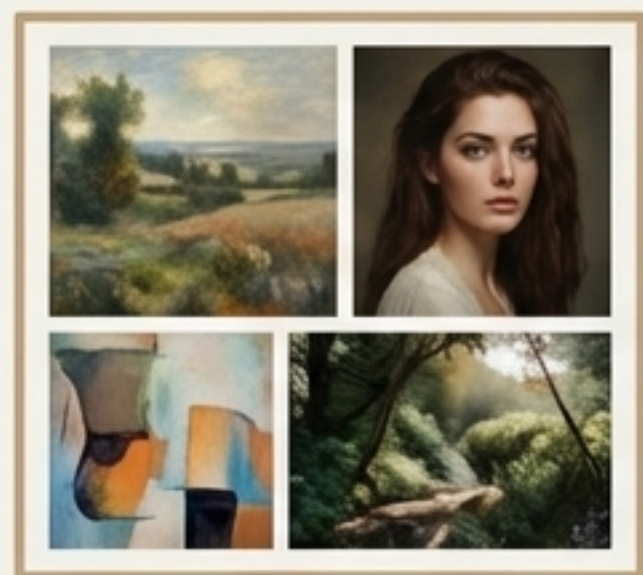
The text prompt acts as a steering wheel. At every step of the denoising journey, the attention mechanism looks at the prompt ("an astronaut on a horse") and adjusts the noise prediction to ensure the emerging image is not just **any** coherent image, but one that specifically matches the text description.



The Quality is Driven by Data and Language

Training Data

The 'aesthetic' quality of Stable Diffusion's output is not accidental. It was trained on datasets like LAION-Aesthetics, which contain images humans have rated as visually pleasing. If trained on logos, it would generate logos.



Aesthetic Training Data

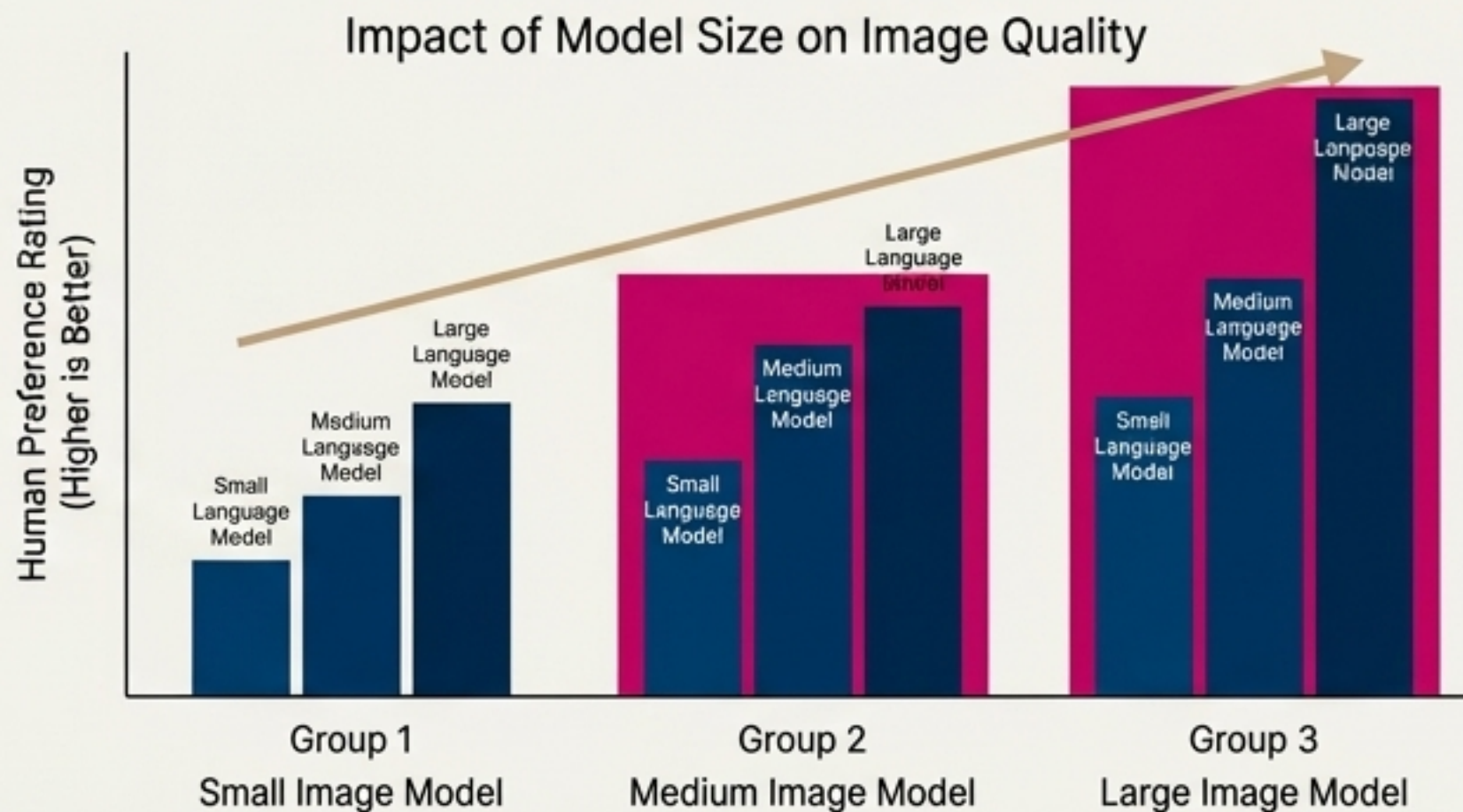


Logo Training Data

The Language Model's Impact

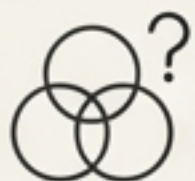
The choice of text encoder is critical. The quality of the generated image is highly correlated with the power of the language model used to interpret the prompt.

A key finding from Google's Imagen paper showed that swapping in larger language models had *more* of an effect on generated image quality than using larger image generation components.

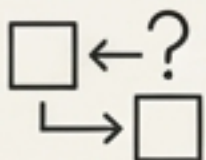


Current Limitations and Difficult Tasks

While powerful, these models still struggle with certain types of tasks. Many of the foundational limitations of CLIP persist in its applications.



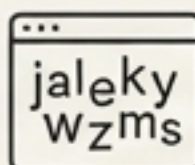
- **Abstract & Systematic Tasks:** Poor at tasks like accurately counting a specific number of objects in an image.



- **Complex Relational Concepts:** Struggles with predicting spatial relationships, like 'the closest car in a photo'.



- **Very Fine-Grained Classification:** Has difficulty distinguishing between specific car models, aircraft variants, or flower species.



- **Text and Typography:** While CLIP has some OCR ability, generating legible and coherent text within images remains a significant challenge.



- **Prompt Sensitivity:** Performance can be highly sensitive to specific wording or phrasing, sometimes requiring 'prompt engineering' to achieve the desired result.

Broader Impacts: Bias, Control, and Responsibility

The ability to design classifiers and generate imagery from language removes the need for task-specific data, but also introduces new challenges related to bias and misuse.

Bias



The way classes are designed heavily influences model biases. When given labels including race and egregious terms like 'criminal', CLIP classifies images of people aged 0-20 in the 'criminal' category at a rate of 32.3%. Simply adding the class 'child' drops this to 8.7%.

Surveillance & Privacy



These models unlock niche tasks with ease. CLIP's performance on celebrity identification (59.2% top-1 accuracy from 100 choices) highlights potential surveillance risks, even if not yet competitive with production models.

These technologies are not just technical tools; they are socio-technical systems. Understanding their capabilities requires understanding their shortcomings, biases, and the new responsibilities they place on their users and creators.